Patents and Industrial Expansion in Nineteenth-Century Britain

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1 Introduction

I want to understand how the rules surrounding invention relate to technological development. Patent records offer a way to explore this question because they connect inventions with arrangements for protection and disclosure. North explains how institutions shape economic incentives [1], while Moser's research on nineteenth-century exhibitions links patent systems with the direction of innovation [2]. These perspectives motivate my comparison of patenting with industrial expansion.

The 1852 patent reform provides the main reference point. Parliamentary debate focused on reducing the expense and difficulty of obtaining patents [6], and the Patent Office took responsibility for registration in October 1852 [7]. A later reform, passed in 1883 and introduced in 1884, included provisions for examining descriptions and handling applications [8].

The project examines changes in recorded patenting around the first reform and compares them with coal production and railway expansion. It also tests whether earlier patent growth helps predict these industrial indicators. The study covers 1802–1902, including the fifty calendar years from 1853 to 1902.

2 Data and historical patterns

Moser's public research collection provides annual patent applications and sealed patents, with its documentation crediting Mitchell's *British Historical Statistics* [3]. The coal and railway observations come from the NBER Macrohistory series available through FRED [4, 5]. Table 1 and Figure 1 show the years available for each indicator.

Table 1: The four indicators cover different parts of the study period.

Indicator	Years covered	Observed years	Internal gaps
Sealed patents	1802–1902	101	0
Patent applications	1852–1902	51	0
Coal output	1854–1902	49	0
New railway mileage	1826–1902	74	3

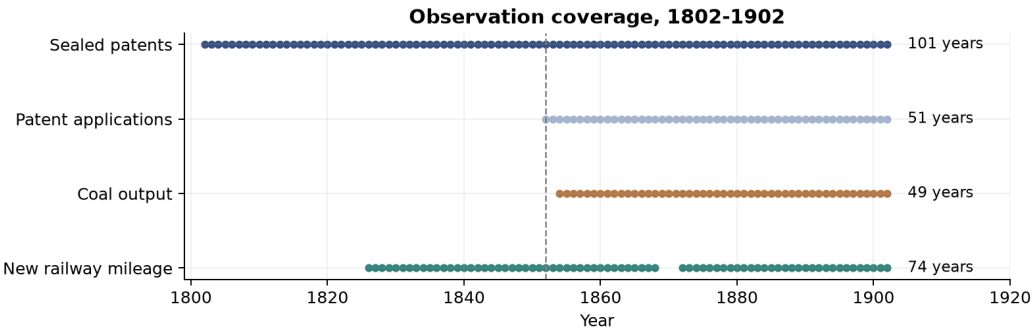


Figure 1: The dots show years with observations. The dashed line marks 1852.

2.1 What the indicators measure

Sealed patents represent completed grants, while applications record an earlier stage of the process. Applications and grants counted in the same year do not necessarily concern the same inventions. Figure 2 therefore shows them separately, and the analysis uses sealed patents as its main patenting measure.

Coal output is measured in millions of long tons and includes Ireland. Railway mileage records the annual increase in lines opened for traffic. The railway data cover Great Britain before 1845 and Great Britain and Ireland during 1845–1912 [4, 5]. The nearby comparisons around 1852 begin in 1845 so that their railway observations cover the same geographical area.

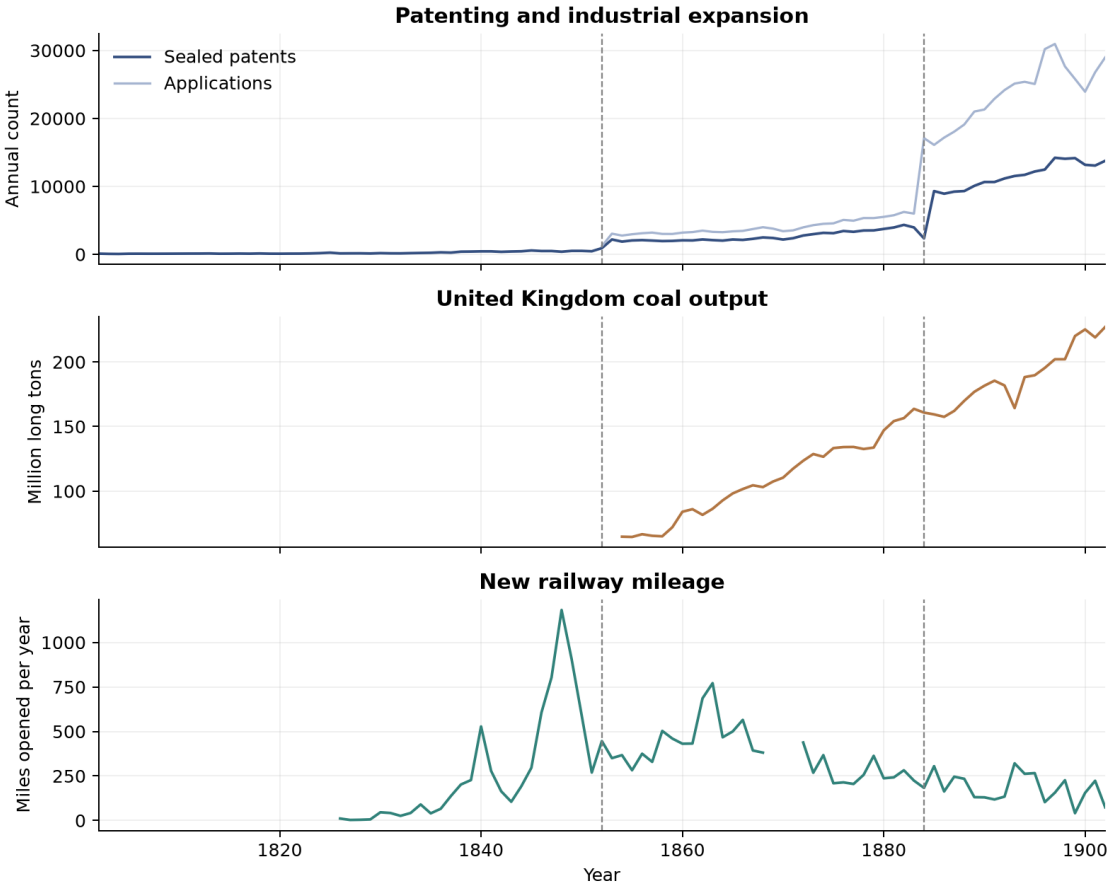


Figure 2: The three panels show patent counts, coal output, and annual new railway mileage. The dashed lines mark 1852 and 1884.

2.2 Patterns over time

The indicators follow different paths. Sealed patents rise sharply in the early 1850s, and coal output increases over much of its recorded period. Annual railway openings fluctuate and later fall from the high construction levels near the middle of the century.

The later reform also coincides with a marked separation between applications and grants. Applications rise from 5,993 in 1883 to 17,100 in 1884. Sealed patents fall from 3,962 to 2,345 over those years, then reach 9,306 in 1885. The annual totals do not explain why the two series move differently.

The railway observations for 1869–1871 remain missing. Annual changes are calculated only when both consecutive years are available. The three main indicators share 46 observed years during 1854–1902, although each comparison uses the observations available for its own variables.

3 Patenting around the 1852 reform

3.1 Growth across different periods

I first compare compound annual growth rates. Each rate expresses the change between a period's first and last values as an equivalent steady yearly rate. The two patent periods in Table 2 each span 49 years between their endpoints, while the two coal periods each span 24 years. Coal output is measured in millions of long tons, and patent figures are annual counts.

Table 2: Patent growth is higher in the later period, while coal growth is lower.

Indicator	Period	Initial value	Final value	Annual growth
Sealed patents	1802–1851	107	455	3.00%
Sealed patents	1853–1902	2,187	13,764	3.83%
Coal output	1854–1878	64.7	132.6	3.03%
Coal output	1878–1902	132.6	227.1	2.27%

The patent growth rate rises from 3.00% in 1802–1851 to 3.83% in 1853–1902. These rates describe growth within each period and leave out the sharp change between 1851 and 1853. Nearby annual averages provide a closer view of that change.

Table 3 compares two seven-year periods on either side of 1852. Mean sealed patents rise from 489.71 to 2,023.43 per year, making the later average 4.13 times the earlier one. Annual new railway mileage falls from 664.29 to 380.86 miles, leaving the later average at 0.57 times the earlier level.

Table 3: Each nearby period contains seven annual observations for both indicators.

Indicator	1845–1851 mean	1853–1859 mean	Later/earlier
Sealed patents per year	489.71	2,023.43	4.13
Railway miles opened per year	664.29	380.86	0.57

Across these seven-year periods, patenting increases while annual railway openings decline. The difference shows why the two indicators need separate interpretation. The increase in recorded grants does not establish a similar increase in railway construction.

Coal output grows at 3.03% per year during 1854–1878 and 2.27% during 1878–1902. The later growth rate is about three-quarters of the earlier rate. Both periods follow the reform. The coal data begin in 1854, so they provide insufficient information to compare coal growth before and after 1852. The available results do not support a general doubling of industrial growth following the reform.

3.2 A closer look at the change in patenting

The next comparison asks whether patenting follows one continuous trend across the reform. I compare a single-trend model with a segmented model that allows both the patent level and its subsequent growth trend to change. The segmented model starts the post-reform period in 1853. It uses 80 observations from 1802–1882 and omits 1852 because registration arrangements changed during that year. The sample ends before the later reform.

Both models are fitted by ordinary least squares using the logarithm of patent counts. An exploratory search also compares possible change years, with at least 15 observations on each side. The Bayesian information criterion (BIC) weighs a closer fit against added model complexity. Lower values indicate a preferred model within this comparison. Choosing the change year adds one parameter to the searched model.

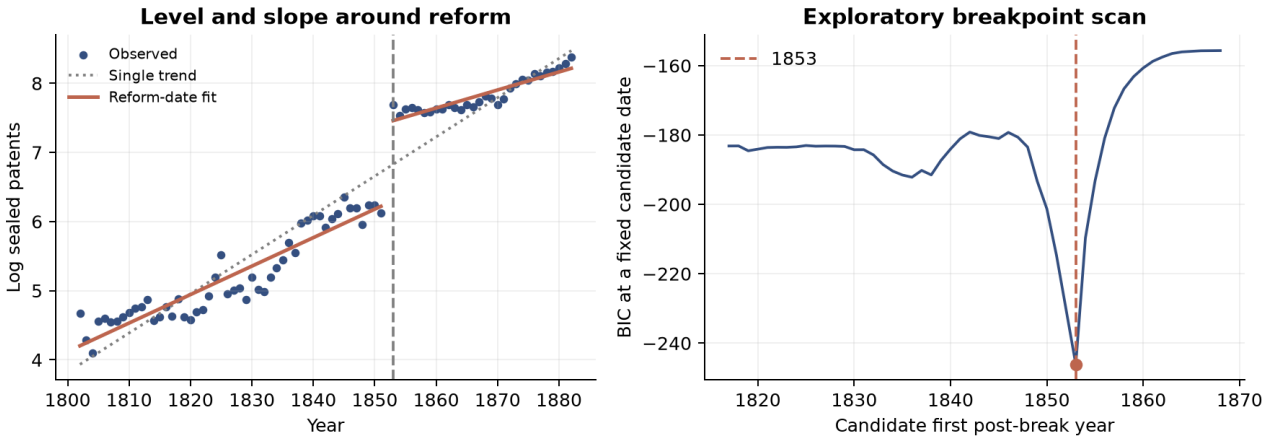


Figure 3: The segmented fit allows the patent level and trend to change. The search selects 1853 as the first observed year after the change.

Table 4: The segmented models describe the 80 patent observations more closely than a single trend.

Model	First post-break year	Parameters	BIC	Log RMSE
Single trend	—	2	−163.49	0.341
Reform date	1853	4	−246.13	0.192
Scanned date	1853	5	−241.75	0.192

Figure 3 and Table 4 show a closer fit when the model includes a break. The root mean squared error (RMSE) on the logarithmic scale falls from 0.341 to 0.192. The search selects the same first post-break year, 1853. Its BIC is higher than that of the fixed-date model because choosing the date adds complexity without improving the fit. Since 1852 is omitted, the model locates the change between the observations for 1851 and 1853.

The estimated increase in the patent level is 218.14%, or about 3.18 times the level implied by continuing the earlier trend into 1853. This estimate differs from the ratio of nearby averages because the model accounts for trends within each period. The fitted annual trend falls from 4.20% before the break to 2.64% afterward. The result combines a sharp upward shift with slower subsequent growth.

The timing is consistent with an institutional explanation, given the reform's focus on patent costs and procedures [6, 7]. Patent totals also depend on decisions to seek protection and on administrative processing. The model describes a change in recorded grants. Its estimates alone cannot establish an equivalent increase in inventions or their adoption.

4 Links with coal production and railway expansion

The next analysis examines correlations during 1854–1902. I compare the series in three ways: their logarithmic levels, their deviations from fitted linear time trends, and their annual logarithmic changes. Coal and patent counts use logarithms; the railway transformation takes the logarithm after adding one to annual openings. These comparisons help distinguish long-term patterns from year-to-year movement.

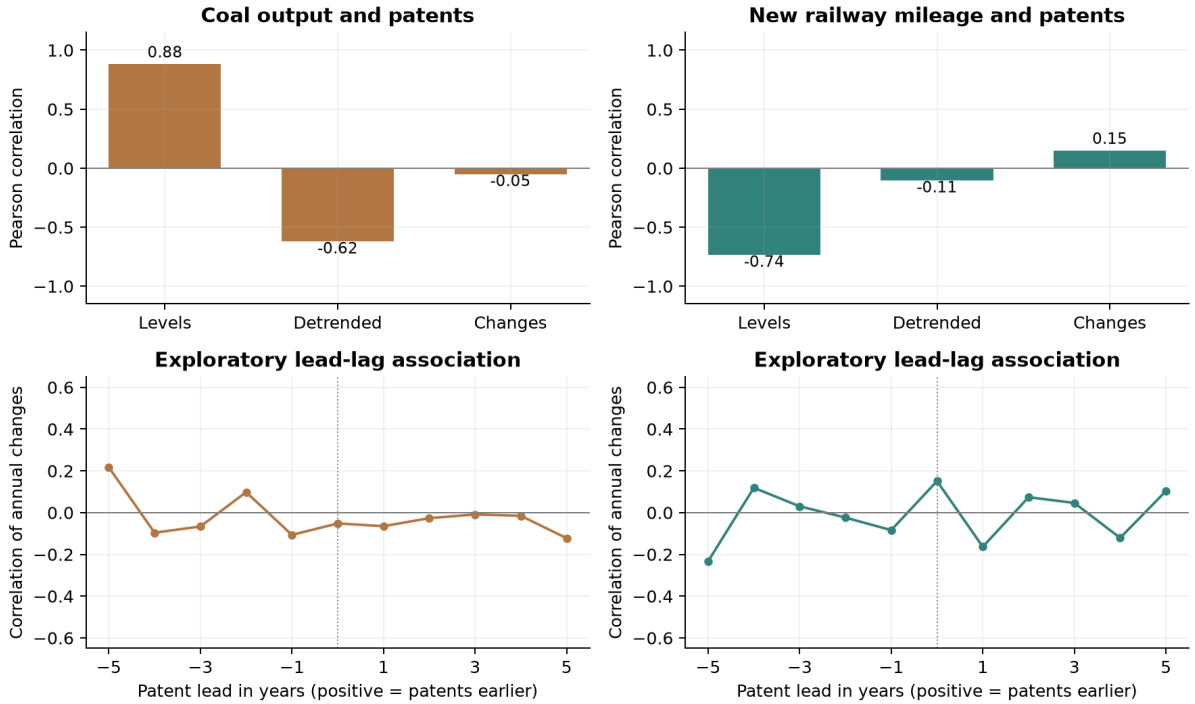


Figure 4: The upper panels compare the three versions of the data. Positive values on the lower panels' horizontal axes place patent changes earlier than industrial changes.

Table 5: The correlations with patenting depend on how the series are compared.

Industrial indicator	Log levels	Detrended levels	Annual changes	Sample sizes
Coal output	0.883	−0.623	−0.052	49 / 49 / 48
New railway mileage	−0.736	−0.105	0.151	46 / 46 / 45

Coal and sealed patents have a strong positive correlation of 0.883 in logarithmic levels. Their annual changes have a correlation of -0.052, while removing a linear trend gives a correlation of -0.623. The positive level correlation therefore does not translate into a clear pattern of annual movement together. The detrended result also depends on the chosen trend model, so its negative value should be interpreted cautiously.

Railway openings have a negative level correlation with patents as construction falls while patenting grows. Their annual-change correlation is 0.151. Table 5 reports different sample sizes because missing observations also affect the availability of annual changes.

The lower panels of Figure 4 compare patent and industrial changes up to five years apart. The coal correlations range from -0.123 to 0.217, and the railway correlations range from -0.233 to 0.151. The profiles show no consistent positive pattern in which patent growth precedes changes in both industrial indicators. These correlations describe the observed relationships, and the analysis does not test their statistical significance across the searched lags.

5 Predictive value and sensitivity

5.1 Does earlier patent growth improve forecasts?

The forecasting experiment asks whether patent growth adds useful information beyond each indicator's recent history. The coal outcome is annual logarithmic growth, measured in log percentage points. The railway outcome is annual newly opened mileage. Their errors therefore retain different units.

A ridge regression provides the baseline, using the previous year's outcome and a linear time trend. A second version adds the previous year's patent growth. Ridge regression discourages large coefficients; the penalty is fixed at 1 and does not apply to the intercept. Predictors are standardized using training data only. A forecast based on the historical mean provides a simple comparison.

The evaluation begins with 20 valid training observations. Each forecast uses earlier years only, and the next training sample includes the newly observed year. Missing current or lagged observations remove that forecast pair. The coal comparison contains 27 forecasts for 1876–1902, and the railway comparison contains 25 for 1878–1902. All three models use the same test years within each outcome.

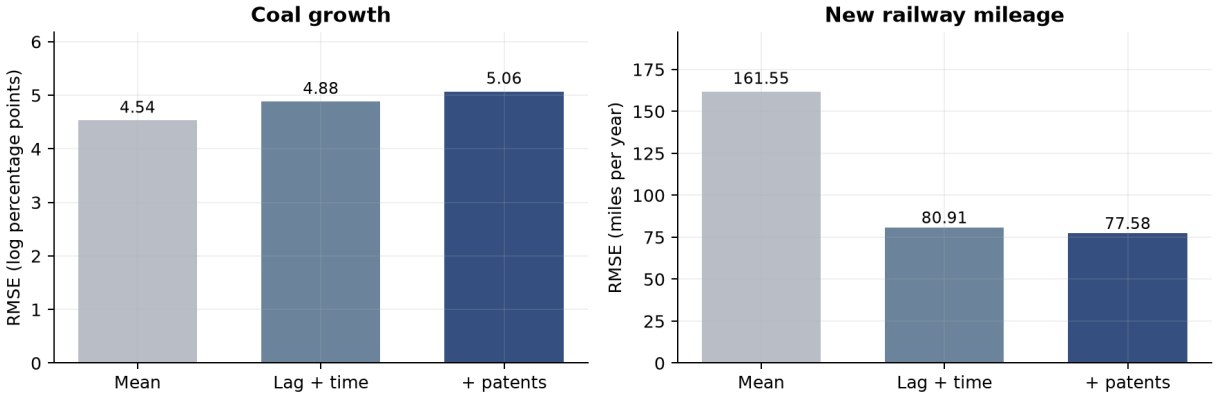


Figure 5: Adding patent growth lowers railway forecast errors and raises coal forecast errors. The panels use different error units.

Table 6: MAE and RMSE summarize the errors in forecasts made from earlier observations.

Target	Model	MAE	RMSE
Coal growth	Historical mean	3.310	4.537
(log percentage points)	Own lag + time	3.353	4.880
	Own lag + time + patents	3.569	5.061
Railway openings	Historical mean	146.52	161.55
(miles per year)	Own lag + time	67.63	80.91
	Own lag + time + patents	62.66	77.58

Figure 5 and Table 6 show different results for the two outcomes. Adding patent growth raises coal RMSE from 4.880 to 5.061, an increase of 3.72%. The historical mean performs better than either coal regression model.

For railway openings, the model using the previous outcome and time trend improves substantially on the historical mean. Adding patent growth lowers RMSE by a further 4.12%, from 80.91 to 77.58 miles per year. Mean absolute error (MAE) also falls from 67.63 to 62.66 miles. This additional improvement is modest, and the experiment does not establish whether it would persist in another period.

5.2 How sensitive are the findings?

The first check varies the initial training sample from 15 to 25 observations. Figure 6 and Table 7 show that adding patent growth consistently raises coal RMSE and lowers railway RMSE across the resulting test periods.

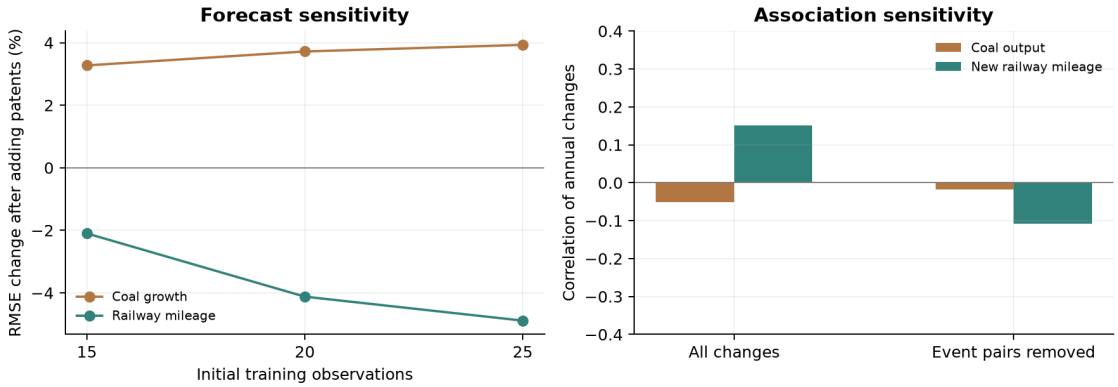


Figure 6: The checks vary training size and exclude change pairs around selected events.

Table 7: The patent feature has the same error pattern across the training samples.

Initial training	Coal test years	Coal RMSE change	Rail test years	Rail RMSE change
15	32	+3.27%	30	−2.10%
20	27	+3.72%	25	−4.12%
25	22	+3.93%	20	−4.89%

The second check excludes changes for 1884–1886 and 1893–1894, covering pairs around the patent transition and the coal-dispute year cited in the source notes [4]. Coal's correlation changes from -0.052 (48 observations) to -0.017 (43). Railway's correlation changes from 0.151 (45 observations) to -0.108 (40). The railway sign change shows sensitivity to the years retained.

6 Conclusion

Recorded patenting rises sharply near the 1852 reform, with about a fourfold increase in nearby annual averages. Coal grows more slowly in the later period, while annual railway openings decline from their earlier peak. Patent growth modestly improves railway forecasts and worsens coal forecasts in the tested models.

The clearest finding concerns recorded patenting. The industrial indicators follow different paths, and their aggregate observations do not trace individual inventions' adoption.

References

[1] D. C. North. "Institutions." *Journal of Economic Perspectives*, 5(1):97–112, 1991. doi:10.1257/jep.5.1.97.

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